NIPPUN SABHARWAL

+1 (447) 902 3085 — nippuns2@illinois.edu — linkedin.com/in/nippun-sabharwal — github.com/nippun-live — Website

SUMMARY

AI researcher focused on world models, foundation model training, and embodied perception. Experienced in designing architectures with Transformers, diffusion, and flow matching, plus building 3D scene representations for sim to real robotics. Proven large scale training and systems optimization, implemented flow matching for a 1B parameter latent autoregressive world model and improved H100 training throughput by 2.3x via mixed precision CUDA kernels.

EDUCATION

University of Illinois Urbana-Champaign Aug 2022 to May 2026

B.S. Computer Engineering *Honors: Samsung Engineering Scholarship, Illinois Engineering Achievement Scholarship*

Coursework: Deep Learning for Computer Vision, Applied Machine Learning, Robotic Manipulation, Parallel Programming

EXPERIENCE

Stealth Robotics Startup, Bay Area Jul 2025 to Aug 2025

Intern, World Models and Hardware *San Francisco, CA*

- Trained a **1B-parameter** latent **flow-matching** world model as a learned simulator, scaling to **0.5B frames** of egocentric (Ego4D) and robot video across **64 H100 GPUs**.
- Designed an action-conditioned latent dynamics stack using a **causal transformer** backbone with a **DiT-style** flow-matching head for long-horizon autoregressive rollouts.
- Improved temporal consistency during rollout by predicting **latent deltas** between consecutive frames rather than full latent frames, reducing compounding drift in autoregression.
- Optimized the training stack with mixed-precision **CUDA** kernels (FP16 intrinsics, tiled GEMM, reductions, concurrent streams), achieving **2.3x** H100 throughput and lower memory use.

National Center for Supercomputing Applications Jan 2025 to Present

Research Assistant (Prof. Narendra Ahuja) *Urbana, IL*

- Built an **unsupervised**, physics-informed pipeline to reconstruct planar scenes from video through turbulent refractive media (water waves, heat shimmer), without clean ground truth.
- Jointly modeled scene geometry and refractive dynamics using ViT-style encoders, NeRF-style encodings, and differentiable wave simulation, enforced via spatiotemporal spectral consistency losses.

AstraZeneca, Global Supply Chain Data Science May 2025 to Aug 2025

Machine Learning Intern *Gaithersburg, MD*

- Built a LangGraph plus RAG agent over millions of supply-chain records, executing multi-step workflows across databases and tools to support allocation and inventory decisions.
- Deployed a secure AWS stack (ECS, ALB, WAF, Secrets, CloudWatch) with GPU-accelerated retrieval (FAISS IVF, float16), achieving sub-**200 ms** time-to-first-token over **10M+** items.

SELECTED PUBLICATIONS

X. Guo, Y. Li, **N. Sabharwal**, S. Sinha, et al. *Toward Engineering AGI, Benchmarking the Engineering Design Capabilities of LLMs.* NEURIPS 2025.

N. Sabharwal, N. Ahuja, "Unsupervised Recovery of Planar Surfaces Under Turbulent Media Using Geometric Reconstruction." *manuscript in preparation for submission to ECCV 2026.*

RESEARCH AND SYSTEMS PROJECTS

3D Reconstruction and Policy Evaluation (ALOHA, UIUC) Aug 2025 to Present

Research Assistant

- Reconstructed a bimanual ALOHA teleoperation setup using **3D Gaussian Splatting** (iPhone LiDAR plus COLMAP), aligned splats to MuJoCo via robust point-to-plane ICP achieving **7 mm RMSE**.
- Built a composite renderer (3DGS background plus MuJoCo robot foreground) at **30 to 60 FPS** across 7+ views, with a LeRobot-compatible observation API for sim-to-real evaluation.

VR Teleoperation and Imitation Learning, UIUC Aug 2024 to Jan 2025

Research Assistant (Prof. Joohyung Kim)

- Built a VR teleoperation interface (Meta Quest 2, IK retargeting) streaming 60 Hz commands and RGBD feedback with **45 to 60 ms** round-trip latency.
- Collected 20k+ demonstrations and trained DAgger plus behavior cloning policies, achieving 80%+ success on randomized pick-and-place in internal evaluations.

RISC-V OS Kernel (C): Sv39 virtual memory, preemptive scheduler, 20+ syscalls, drivers (VirtIO, UART, RTC, PLIC), crash-safe WAL file system.

SKILLS

ML and training: Transformers, diffusion, flow matching, VAEs, action-conditioned sequence modeling, mixed precision, profiling and optimization, multi-node GPU training.

Languages: Python, C, C++, CUDA, Java, SQL, Verilog, SystemVerilog, RISC-V assembly.

Frameworks and tools: PyTorch, HuggingFace, Linux, Git, Docker, Kubernetes, AWS.

3D and vision: NeRF, 3D Gaussian Splatting, COLMAP, PyTorch3D, Open3D, OpenGL, OpenCV.